

AvatarFit Home Workout Survey

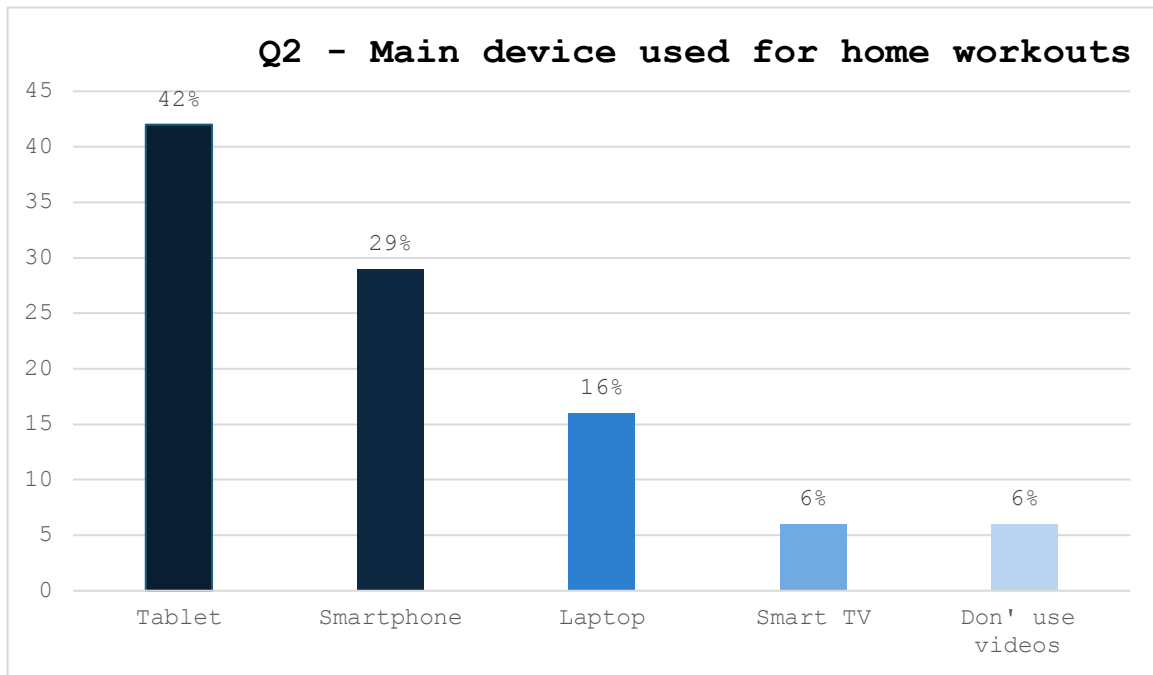
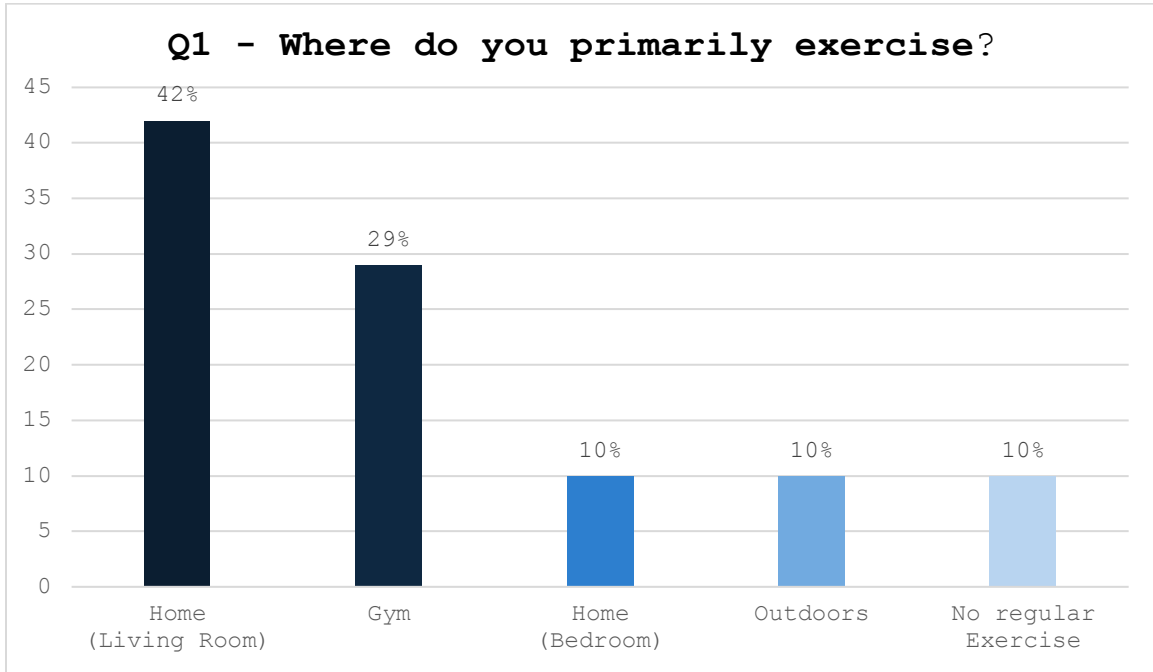
Key Findings Report

HCIW Evaluation | n= 31 respondents

31 total responses	58% avoided an exercise <i>due to injury fear</i>	84% uncomfortable with webcam <i>in live class with strangers</i>	77% prefer neon avatar <i>over real video</i>
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1. Context: Where and How People Exercise

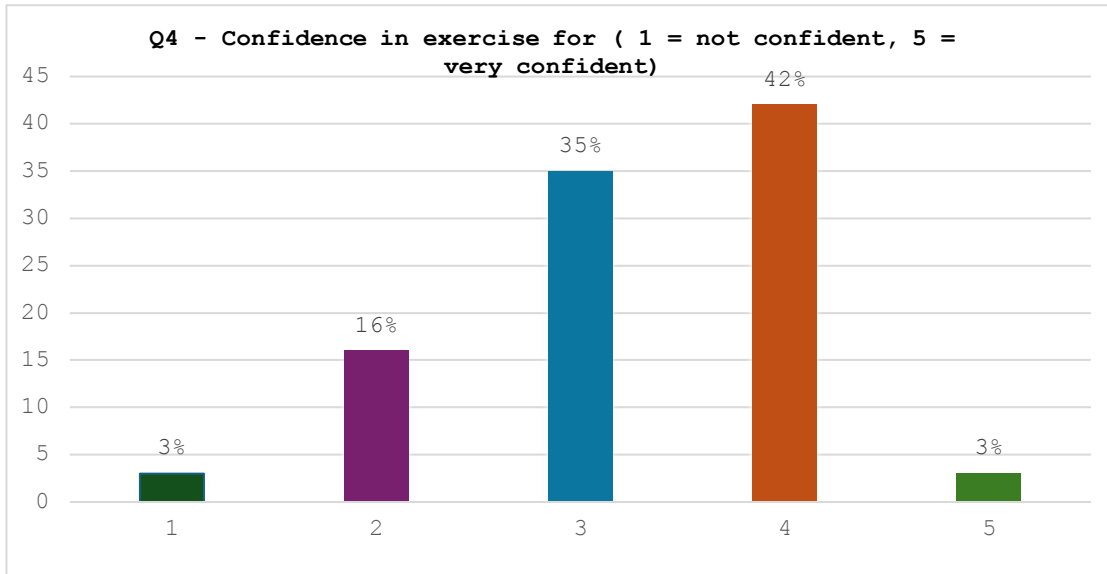
Q1–Q3 establish the physical context in which AvatarFit would be used.



52% of respondents exercise at home (living room + bedroom). Most use a tablet or smartphone at 1–2 m from the screen — this constrains UI design: feedback must be readable from a distance.

2. The Problem: Form Confidence and Injury Fear

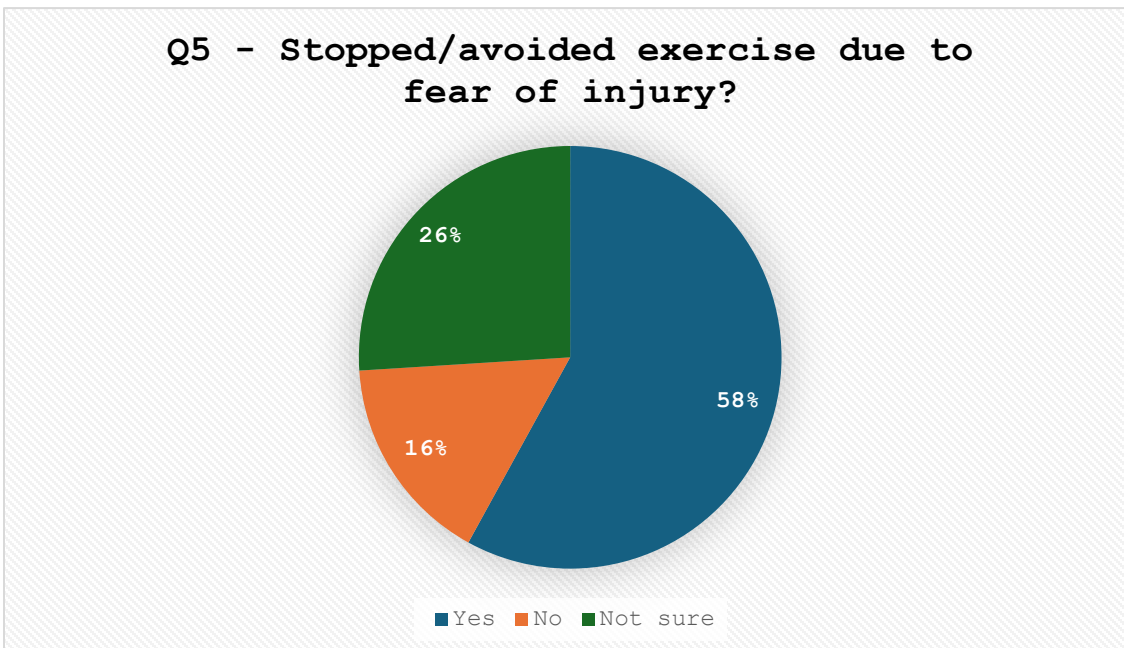
Q4–Q5 measure how confident users feel about their exercise form and whether injury concerns affect their behaviour.



58% avoided exercise due to injury fear: of respondents

54% rate from confidence at 3 or below: (out of 5)

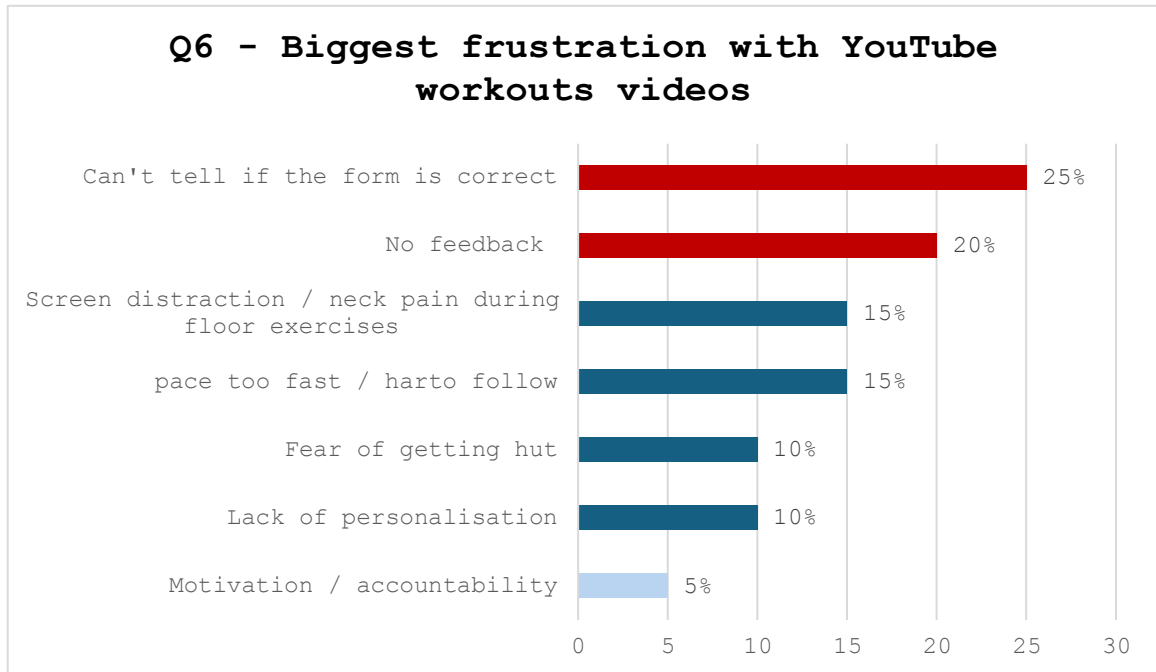
Only 3% are fully confident in their form:



More than half of users are not confident in their form, and the majority have modified their behaviour due to injury fear — confirming a real, felt problem that AvatarFit directly addresses.

3. Frustrations with Pre-Recorded Workout Videos

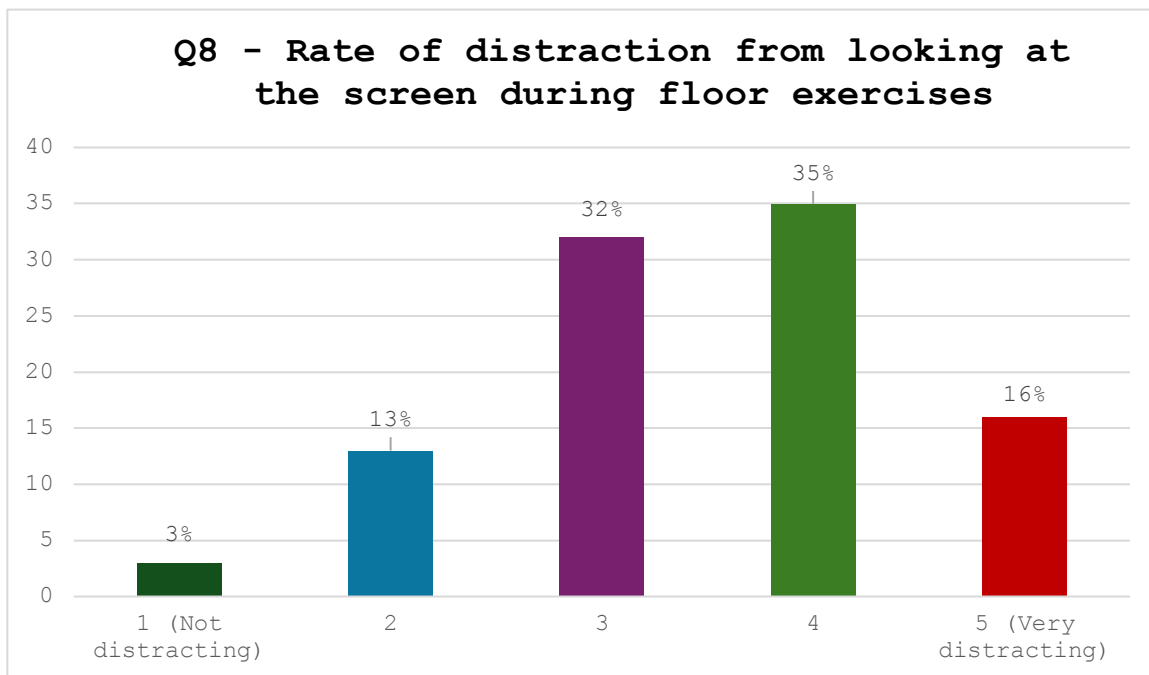
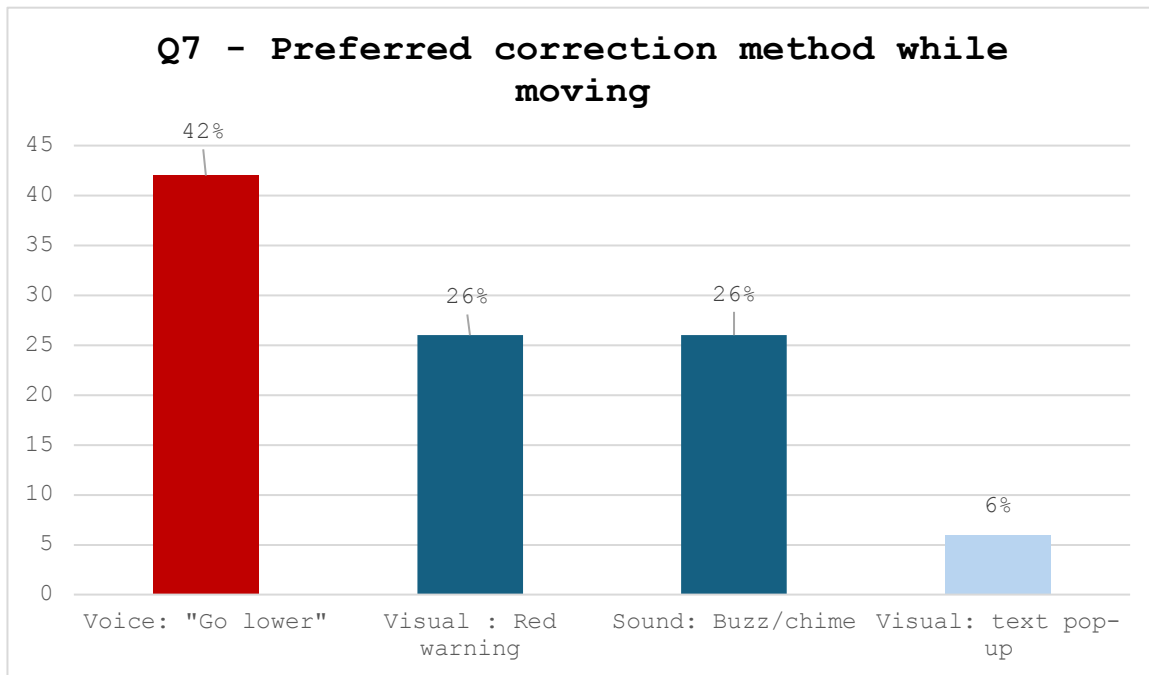
Q6 was open-ended. Responses were grouped by theme (n = 31 answered, some left blank).



The top frustrations all converge on one root cause: the absence of real-time, personalised feedback. Users do not know if they are doing exercises correctly, and there is no mechanism to tell them.

4. Preferred Feedback Method

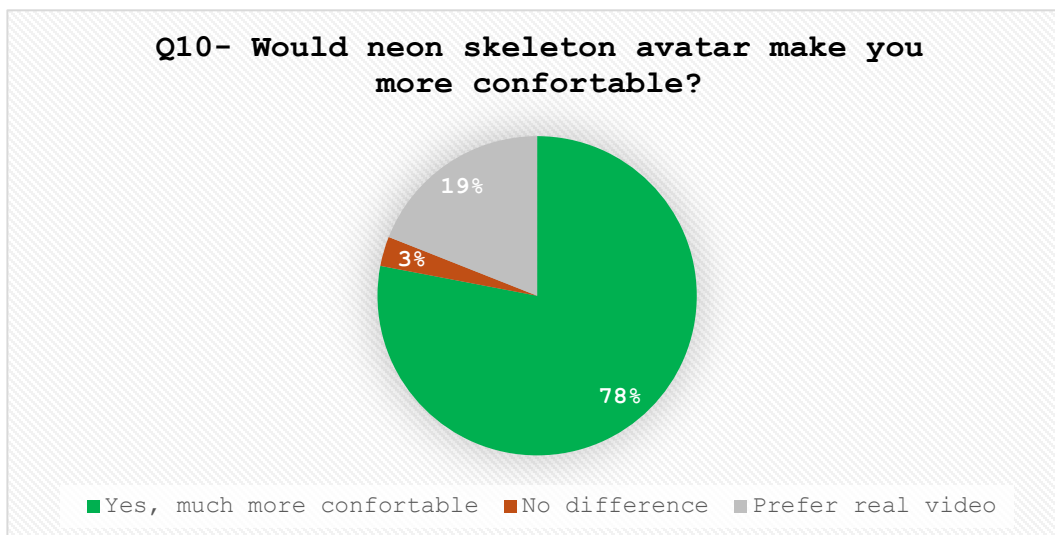
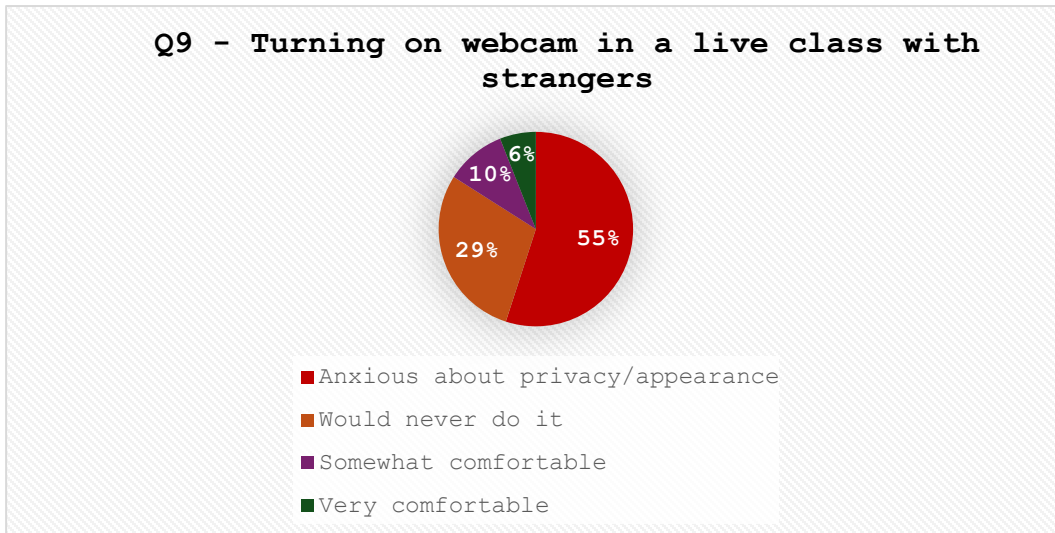
Q7 asks which type of correction users prefer while moving.



Voice feedback is the top individual preference (42%), but visual + audio options together account for the majority. 83% find floor-exercise screen-looking moderately or very distracting — suggesting audio cues are especially important for those movements.

5. Webcam Privacy and Avatar Acceptance

Q9–Q10 address one of AvatarFit's core design decisions: using a neon skeleton avatar instead of real video .



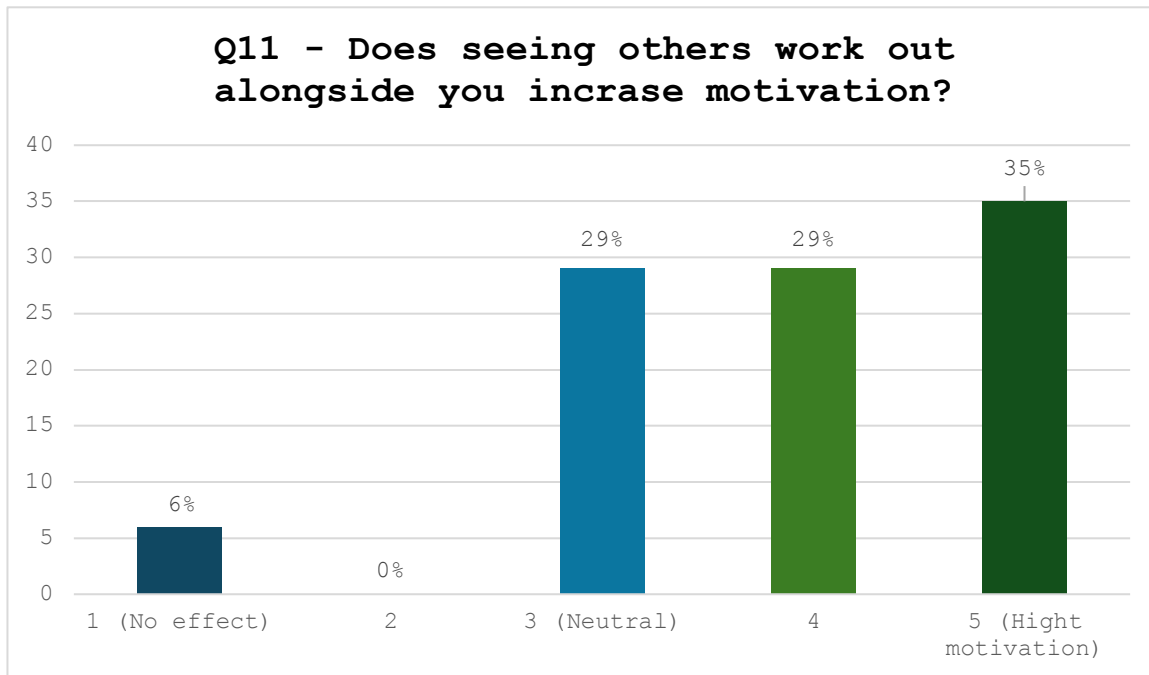
84% uncomfortable with real webcam: (anxious or would never do it)

77% would prefer the neon skeleton avatar:

These results directly validate AvatarFit's avatar-based privacy design. The vast majority of users are uncomfortable sharing real video — and the skeleton avatar resolves this concern for 77% of them.

6. Social Motivation: Working Out Alongside Others

Q11 tests whether the social/group workout concept increases motivation (n = 31).



64% rated 4 or 5 — social presence, even via AI avatars, meaningfully increases motivation. This supports the "Just Dance for fitness" social workout concept at the core of AvatarFit.

7. Formal HCI Problem Statement

Based on the survey findings, the HCI problem addressed by AvatarFit is formally stated as follows:

People who exercise at home cannot reliably verify whether their posture and form are correct in real time. Without expert oversight, they risk reinforcing bad habits or sustaining preventable injuries. Existing solutions — primarily pre-recorded video tutorials — are one-directional and provide no personalised feedback. Users must frequently interrupt their workout to check the screen, which itself becomes a source of physical discomfort. Additionally, privacy concerns prevent users from engaging with live remote fitness classes that would otherwise provide the interactivity they need.

Survey evidence (n = 31):

- 58%** have stopped or avoided an exercise due to fear of injury
- 54%** rate their form confidence at 3 or below (scale of 1–5)
- 83%** find looking at a screen during floor exercises moderately or very distracting
- 84%** are uncomfortable or unwilling to turn on their webcam in a live class
- 77%** would feel more comfortable using an anonymous skeleton avatar
- 64%** report that social presence (even AI avatars) increases their motivation

Proposed solution

AvatarFit is a browser-based fitness application that uses real-time pose detection (MediaPipe) to provide immediate, personalised form feedback during exercise. Users are represented as anonymous neon skeleton avatars, removing the privacy barrier identified in the survey. The application gives visual and audio feedback on joint angles in real time, addressing the core problem of unverified form during home workouts.